BANKING AI ASSISTANT FOR UNSIGHTED PEOPLE USING NLP

A Project report submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
INFORMATION TECHNOLOGY

Submitted by

K.Abhiram	**319126511095**
P.Hanisha	**319126511111**
M.Eeshvar Gopal	**319126511098**
K.Likhita	**319126511097**

Under the guidance of
Mr. G Pandit Samuel
Assistant Professor

PRAGNANAM BRAHMA
ANITS

DEPARTMENT OF INFORMATION TECHNOLOGY

ANIL NEERUKONDA INSTITUTE OF TECHNOLOGY AND SCIENCES (AUTONOMOUS)
(Permanent Affiliation By Andhra University & Approved by AICTE Accredited by NBA (ECE, EEE, CSE, IT, Mech, Civil & Chemical) & NAAC)
Sangivalasa, Bheemili Mandal, Visakhapatnam dist.(A.P)

2022-2023

DEPARTMENT OF INFORMATION TECHNOLOGY

ANIL NEERUKONDA INSTITUTE OF TECHNOLOGY AND SCIENCES (AUTONOMOUS)

(*Permanent Affiliation by Andhra University & Approved by AICTE Accredited by NBA (ECE, EEE, CSE, IT, Mech. Civil & Chemical) & NAAC*)

Sangivalasa, Bheemili Mandal, Visakhapatnam dist.(A.P)

CERTIFICATE

This is to certify that the project report entitled **"Banking AI assistant for Unsighted people using NLP"** submitted by **K.Abhiram (319126511095), P.Hanisha (319126511111), M.Eeshvar Gopal (3191265111098), K.Likhita (3191265111097),** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Information Technology** of Anil Neerukonda Institute of technology and sciences, Visakhapatnam is a record of bonafide work carried out under my guidance and supervision.

Project Guide
Mr. G Pandit Samuel
Assistant Professor
Department of IT
ANITS

Head of the Department
Dr. M. Rekha Sundari
Professor
Department of IT
ANITS

ACKNOWLEDGEMENT

We would like to express our deep gratitude to our project guide **Mr. G Pandit Samuel** Assistant Professor, Department of Information Technology, ANITS, for his/her guidance with unsurpassed knowledge and immense encouragement. We are grateful to **Prof. M.Rekha Sundari**, Head of the Department, Information Technology, for providing us with the required facilities for the completion of the project work.

We are very much thankful to the **Principal and Management, ANITS, Sangivalasa,** for their encouragement and cooperation to carry out this work.

We express our thanks to all **teaching faculty** of the Department of IT, whose suggestions during reviews helped us in accomplishment of our project. We would like to thank **all non-teaching staff** of the Department of IT, ANITS, for providing great assistance in the accomplishment of our project.

We would like to thank our parents, friends, and classmates for their encouragement throughout our project period. At last but not the least, we thank everyone for supporting us directly or indirectly in completing this project successfully.

PROJECT STUDENTS

K. Abhiram(319126511095)

P. Hanisha(319126511111)

M. Eeshvar Gopal(319126511098)

K. Likhita(319126511097)

DECLARATION

We hereby declare that the project work entitled **"Banking AI assistant for Unsighted people using NLP"** submitted to the Anil Neerukonda Institute of Technology and Sciences is a record of an original work done by **K.Abhiram (319126511095), P.Hanisha (319126511111), M.Eeshvar Gopal (3191265111098), K.Likhita (3191265111097)** under the esteemed guidance of **Mr. G Pandit Samuel,** Designation of Information Technology, Anil Neerukonda Institute of Technology and Sciences, and this project work is submitted in the partial fulfillment of the requirements for the award of degree **Bachelor of Technology** in **Information Technology.** This entire project is done with the best of our knowledge and has not been submitted for the award of any other degree in any other universities.

K. Abhiram(319126511095)

P.Hanisha(319126511111)

M.Eeshvar Gopal(319126511098)

K.Likhita(319126511097)

ABSTRACT

The main objective of banking is to provide the services to the people to manage their money in a safe way. Banking functionalities include “Accepting deposits of money from the public for Lending or Investment, repayable on demand and can be withdrawn by cheque, draft, or any other means . when it comes to the blind people they must depend on others for daily activities. So, we are trying to help the blind in the banking sector which enables them to do physical activities like filling a form, check balance, update details etc. on their own. People can use the banking services such as deposit, withdraw, loans, etc on their own. But the scenario when it comes to blind people, he/she must have to depend on others to operate the bank account on behalf of the blind person. The bank may accept visually impaired customer’s requests to appoint a person or multiple persons as their Power of Attorney or a Holder to operate their bank account. If the person is not the trusted one, then the blind people may lose their power on their account where he/she may lose the money or information which is private to his account. We have come up with an innovative solution to solve all the above-discussed problems.

Banking AI assistant for unsighted people using NLP is an application where the user is provided with the banking services to operate on their own using advancement of technology. The application works in such a way that the people can easily operate their banks just by speaking their details and access the services that are used by the normal persons. Besides this the operations are performed in a secured way that is by face detection, recognizing the finger pattern(or any other biometrics in the future), detection of more than one face by the system then it will warn and halt the operations. When a user comes to operate their bank this system will take their voice as input through speaker and give the required result for the user. After taking the voice input the system will ask the questions for the operations such as form filling, banking services. And after form filling then it will print the form by authentication with digital signature(fingerprint authentication). By using this system, the third person's involvement between the visually impaired person and the banking system will be removed and the banking operations are more secure.

The Rule-based is the simple one when compared to other ones and can respond to customers based on predefined original, human-generated rules. Communication in this form tends to be text-based and often uses on-screen buttons, giving the user a wide choice of responses. These bots can also work by understanding certain keywords and phrases in written communication.

INDEX

Chapter Name **Page No.**

LIST OF FIGURES

CHAPTER 1

Chapter 1
INTRODUCTION

1.1 Introduction

What is a chatbot?

A chatbot is a computer software powered by artificial intelligence that can mimic a conversation with a human user. These programs are designed to interact with users in a way that mimics human conversation, providing a more intuitive and natural way to engage with technology. Chatbots useNLP & ML algorithms to understand and interpret the user's input & respond with relevant information or actions. They can be used for a wide range of applications, from customer service to e-commerce, education, and healthcare. Chatbots are becoming increasingly popular due to their ability to provide personalized, real-time support, improve customer engagement, and reduce operational costs. With the advancements in natural language processing and artificial intelligence, chatbots are becoming more sophisticated and capable of handling complex conversations, making them an indispensable tool for businesses and individuals alike.

A chatbot is a program that is designed to simulate human conversation, using NLP and artificial intelligence to interact with users. Chatbots can be programmed to respond to text or voice commands, and can be used for a variety of purposes, such as customer service, virtual assistants, education, healthcare, and more. They are able to understand and interpret human language, and can provide relevant information, answer questions, and even carry out tasks or transactions. Chatbots are becoming increasingly popular due to their ability to provide real-time support, improve efficiency, and reduce costs, making them an indispensable tool.

Need for a Banking Bot

Banking bots can be extremely helpful for visually impaired people, as they provide an accessible and convenient way to manage their finances independently. Here are some ways that a banking bot can help visually impaired people:

- Voice recognition: Banking bots are equipped with voice recognition technology, which allows visually impaired users to interact with the bot using their voice. They can ask questions, give commands, and carry out banking transactions without needing to see a screen or type.

- Text-to-speech capabilities: Banking bots can also read out the information on the screen, such as account balances, transaction history, and other account details, using text-to-speech capabilities. This makes it easier for visually impaired users to access the information they need.

- Intuitive interface: Banking bots are designed with an intuitive interface that is easy to navigate using simple voice commands. This means that visually impaired users can perform a range of banking transactions without needing to rely on sighted assistance or visit a physical bank branch.

- Availability: Banking bots are available , which means that visually impaired users can access their accounts and perform banking transactions at any time, from the respective bank, without needing to rely on sighted assistance.

Banking bots are such innovations that have revolutionized the way we manage our finances. A banking bot is a chatbot that uses NLP & AI to communicate with customers, providing them with a range of banking services and support. This technology is transforming the way people access and manage their finances, providing them with a more personalized, convenient, and efficient experience. With a banking bot, customers can check their account balances, transfer funds, pay bills, and perform other banking transactions.

Overall, banking bots provide a more accessible and convenient way for visually impaired people to manage their finances independently, improving their financial inclusion and independence.

1.2 Motivation For Problem

Blind people face several challenges when it comes to the banking sector due to their visual impairment. Here are some of the problems that they might encounter:

- Accessing information: Blind people may have difficulty accessing information about their account balances, transaction history, and other important account details, especially if it's only available in a visual format.

- Security concerns: Blind people may face challenges in keeping their bank accounts secure, as they may not be able to read written security information, such as PIN numbers and security codes.

- Physical accessibility: Visiting a bank branch in person may be difficult for blind people, as they may face obstacles in finding their way around the branch, using ATM machines, and completing transactions independently.

- Lack of assistance: Blind people may require assistance from a third party, such as a family member or friend, to access their bank accounts, which can be inconvenient and compromise their privacy.

- Limited financial products: Blind people may have limited access to financial products and services, such as online banking or mobile banking applications, which are not designed to accommodate their needs.

Overall, these challenges can make banking more difficult and less accessible for blind people. However, with the development of new technologies, such as banking bots and other assistive devices, it's becoming easier for visually impaired people to manage their finances independently.

As the idea is to make banking available to the blind without any external support, the automation of the banking can be done by the system which can take a vocal input and process the input in all the ways possible according to the user requirement and also must be able to give out the apt response. This system should also be able to do the major and minor operations involved and hence to fulfill all the needs we propose "Banking AI assistant for Unsighted people using NLP".

1.3 Problem Statement

Nowadays the banking sector is growing in all kinds of technical stuff. But the utility of the banking services for the blind people has always been complex.Visually impaired people face several challenges when it comes to banking, such as difficulty accessing account information, security concerns, physical accessibility, and a lack of assistance. These challenges can make it challenging for them to manage their finances independently, resulting in inconvenience, privacy issues, and a lack of financial empowerment. A solution is needed to address these challenges and create a more inclusive banking experience for visually impaired people. So this project's base idea is to make an automatic bot designed specifically for the blind to interact with the system through voice commands allowing them to easily access account information, complete transactions, and receive personalized assistance. As the project involves banking, the first thing to implement is the security to use this application. So considering it, at different stages of the usage of the application, several security steps are implemented. As a part of it the user will be identified by the system and then the later part ie. The filling of the sheet will be done automatically using the user's data in the bank server and also by the input of the user. This solution has the potential to improve the financial independence and overall quality of life for visually impaired people.

Introducing our innovative banking bot, specifically designed to provide seamless banking experience to the visually impaired. This bot is equipped with state-of-the-art technologies that make it easy for users to access and manage their bank accounts without any assistance. The bot's intuitive interface, voice recognition and text-to-speech capabilities ensure that users can interact with it easily and efficiently. With our bot, blind users can perform a wide range of banking transactions, from checking account balances & transaction history, to transferring funds and paying bills, all without the need for assistance from a third party. Our banking bot is a game-changer for the visually impaired, empowering them with the ability to manage their finances with ease and independence.

1.4 Background Information

1.4.1 Historical development prior to the project

ELIZA is one of the earliest known chatbots, developed in the mid-1960s by Joseph Weizenbaum, a computer scientist at MIT. The name "ELIZA" is actually a reference to the character Eliza Doolittle from George Bernard Shaw's play "Pygmalion."
ELIZA was designed to simulate a conversation with a psychotherapist, and it achieved this by using natural language processing techniques to respond to user input. The program analyzed the user's input and generated a response that reflected back some of the user's own words or phrases, creating the illusion of a conversation.
Although ELIZA was not capable of true artificial intelligence, it was groundbreaking for its time and sparked interest in natural language processing and chatbot development. Today, chatbots have become much more sophisticated and are used in a variety of applications, from customer service to personal assistants. ELIZA remains an important milestone in the history of chatbot development and continues to be studied and analyzed by researchers in the field of artificial intelligence.

1.4.2 Project Field

This project belongs to the domain of AI based chatbot and NLP. The project involves the use of advanced technologies such as NLP , speech recognition, and ML to create a chatbot that can interact with visually impaired users in a way that is accessible and user-friendly. Forms are filled by the chatbot by recognising the speech of the user.For this purpose we need open CV , speech recognition . Ai protected security is implemented.

1.4.3 Braille Education System

Author: Louis Braille. Year: 2016 Theoretical Model: The current blind education system is used by the Braille education system invented by Louis Blair. In this method, each alphabet is represented by a raised point in the embossed paper. The blind can read the letters by touching the marked points. The percentage of blind people who are literate in Braille in developing countries like Pakistan is alarmingly low, even though Braille is a necessity for all blind people and the only means for blind children to learn to read and write. In schools in developing countries, less than 3% of visually impaired children are presently learning Braille. Despite studies demonstrating that 80% of visually impaired people with jobs can read and write Braille with ease, this still occurs. Braille literacy is therefore essential for obtaining work and fully participating in society. This research paper describes the creation of a portable, self-learning, low-cost, low-power, and user-friendly Braille system.

CHAPTER 2

Chapter 2
LITERATURE SURVEY

2.1 Introduction:

The principal purpose of this is to provide a literature analysis of inventory management articles published in large logistics companies, identify topics from the literature, and provide future directions for inventory management research. A literature review identifies, evaluates, and synthesizes literature relevant to a particular research area. Finally, we summarize the literature review on inventory management systems and explain why this system is essential for all organizations.

2.2 Literature Review:

For the Visually impaired people, in the banking sector, there is no proposed system which can automate the banking processes with enough security. Hence in order to automate the process with a friendly user interface, we propose our system for various banking purposes, which will also let us remove the necessity of the requirement of the other person along with the blind people.The proposed Banking AI Assistant is a tool where the user is enabled to do all kinds of bank transactions all by himself without any other's help. Here as the user is visually impaired, the proposed system will interact with the user only through the output of voice commands. The only thing any user has to do in advance is to remember his account number and other primary details. The rest of the part of the operations are done by the system. The purposes of the system are categorized into 2 types. Some smaller operations are done by minor segments of the system, and major operations are done with some extra authentication.

[1] A Smart Personal AI Assistant for Visually Impaired People by Shubham Melvin Felix, Sumer Kumar, and A. Veeramuthu in A Smart Personal AI Assistant for Visually Impaired People. In their research, they demonstrated how this suggested system can recognise the user who is engaging with it. a mechanism allowing vision challenged persons to recognise objects. Both the visually challenged person and society will benefit from this system. Additionally, it aids in determining the direction of greatest brightness and primary colors. For object recognition, a hybrid approach that combines Euclidean Distance measurements and Artificial Neural Networks is presented. The video that was recorded using a camera will be divided into several frames, and each frame will be compared to the preceding frame in order to save data and provide answers based on that data.

[2] Humorist Bot: Bringing Computational Humor in a Chat-Bot System by S. J. du Preez & S. Sinha's paper AN INTELLIGENT WEB-BASED VOICE CHAT BOT,

discussed the usage of the Intelligent voice chatbot. As the proposed system requires the functionality of being able to interact with the user by using a voice exchange. Some essential features are to be taken from the paper.The first demonstration of this article was developed in Java, and the applet was inserted using HTML. The website is hosted by Apache and has the applet integrated in it. Open source database management software is used to maintain the database and website. In the event that an XML message is invalid, the interface also offers fault response creation. The bound object is then used to encapsulate the error location in XML,which is then communicated over SOAP. In the event that the requirements or message format change, the interface will be generated immediately.

[3] Enabling Finger-Touch-Based Mobile User Authentication via Physical Vibrations on IoT Devices by Xin Yang,Song Yang,Jian Liu,. His method gives our project the modules it needs to carry out fingerprint authentication while also enabling an Internet of Things device to identify a fingerprint and carry out a verification using data from a database. This technique uses low-cost physical vibration to confirm finger input on commonly touched surfaces. Our approach broadens the applicability of digit input verification. It provides an inexpensive, practical, and better security solution by fusing passcode, behavioral and physiological characteristics, and interface dependencies, in contrast to passcode- or biometrics-based solutions. Vibration signals that function on a variety of surfaces are combined with touch recognition in the suggested technique. Three separate passcode secrets—a PIN, lock pattern, and basic motions—as well as novel methods, are enabled to differentiate between fine-grained finger inputs. These methods record both cognitive and physiological factors, such as contacting area, touching power, and others, by eliminating particular elements from the frequency domain. One set of inexpensive, compact, and easy-to-connect vibration generators and sensors is used for the method. (e.g., a door panel, a stovetop or an appliance). Numerous studies show how well our system performs in terms of user authentication accuracy (over 97% in just two attempts), false positive rate (less than 2%), and resilience to a variety of attacks.

[4] Speech-based Virtual Travel Assistant for Visually Impaired by N. Sripriya, S.Poornima,S.Mohanavalli, R.Pooja Bhaiya,V.Nikita in SpeechBased Virtual Travel Assistant For Visually Impaired explored the application of speech bots in numerous technological domains. A speech bot is a kind of chatbot that offers users seamless and highly customized assistance. Speech bots can help you organize a variety of aspects of your life through human-like interactions, like sending reminders, making appointments, and buying tickets, among other things. This increases productivity throughout your firm, lowers the possibility of human error, and enhances overall efficiency. These bots accept voice input from users and create a conversational interface using natural language processing. Following that, the bots respond to the queries with relevant information. A chatbot is made to understand human language, reply to it like a human, and gradually adopt more kind behavior. Well-known examples of bots are Siri and Google Assistant.

[5] A Virtual Assistant for Visually Impaired Persons by Md.Rakibuz Sultan & Md Moinul Hoque in ABYS(Always By Your Side): A Virtual Assistant for Visually Impaired Persons proposed an approach in building a model useful for a virtual assistant for the impaired persons. The application is not user-dependent, and human voice commands are spoken in English to the application. The user can manage installed programmes, send and receive emails, or browse the internet using this application model. The model enables users to access the programme via voice commands and is also capable of browsing a news portal.

[6] Assistant for Visually Impaired using Computer Vision by Pratik Vyavahare & Syed Habeeb in Assistant for Visually Impaired using Computer Vision discussed the overview on how to create a virtual assistant using the various technologies in all aspects. In order to explain the functions of a narrative to the user, a virtual assistant for the blind is constructed employing a variety of technologies, including computer vision, object recognition, and text-to-speech. By translating the scenes that the user needs to text that describes the key amenities in audio format, we create an audio story.

[7] Speech to text conversion for multilingual languages by Yogita H. Ghadage, Sushama D. Shelke in Speech to Text Conversion for Multilingual Languages talked about converting languages based on user preference. In the current study, a multilingual speech-to-text conversion system is described. Speech signals contain information that serves as the basis for conversion. The most important and organic form of communication for humans is speech. A speech-to-text system requires a string of words as the output after receiving a human-spoken phrase as its input. The aim of this system is to gather, characterize, and identify information about speech. Cepstral Mel-Freq. The proposed system is built using Support Vector Machine, Minimum Distance Classifier, and Coefficient Feature Extraction techniques. A database is stored with pre-recorded speech utterances. Most of the database is used for training and testing. On samples taken from the training database, the training phase is applied, and features are subsequently retrieved. The features of each sample are combined, then each feature vector is saved as a reference. A sample from the testing portion is provided to the system, and its features are extracted. The words that are produced have the maximum degree of similarity between these attributes and the reference feature vector according to a calculation. The system is built using the MATLAB programming environment.

CHAPTER 3

CHAPTER 3
SYSTEM ANALYSIS

System analysis is a very complex process used by professionals in order to maintain and develop computer-based systems. It helps to understand what a system does and what are its requirements and how it will be maintained, so it's quite crucial to know and understand the user's requirements.

The design part for BANKING AI ASSISTANT FOR UNSIGHTED PEOPLE system is divided into sections

3.1 Software requirement specifications:

3.1.1 Python:

Guido Van Rossum's Python is an interpreted, high-level, general-purpose programming language. It was first introduced in 1991, and its design philosophy places a strong emphasis on code readability through the noticeable use of whitespace. Python's large standard library, which offers a variety of tools and modules for jobs including web development, scientific computing, data analysis, and more, is one of its main advantages.Python also has a sizable and vibrant development community that contributes to a huge ecosystem of third-party packages and libraries, further extending its functionalities. Its language constructs and object-oriented methodology are designed to aid programmers in creating clean, comprehensible code for both little and big projects. Python has rubbish collection and dynamic typing. Python is a great language for novices thanks to its simplicity and usability, yet experienced programmers also like it because of its strength and versatility. Its ability to integrate with other languages and systems also makes it a valuable tool for building complex software applications. Procedural, object-oriented, and functional programming are just a few of the programming paradigms it supports.

3.1.2 PIP:

Python packages are installed and managed using PIP (Python Package Installer), a command-line utility and package management system. It is the standard package manager for Python and is included by default with most Python installations. With PIP, users can install, update, and remove Python packages, as well as manage dependencies between packages. PIP is a crucial tool for Python developers and users, as it simplifies the process of installing, updating, and managing Python packages, making it easier to work with third-party libraries and dependencies.

3.1.3 Speech Recognition:

With the aid of speech recognition technology, a computer can comprehend spoken words. It is sometimes referred to as speech-to-text or automated speech recognition (ASR) (STT). In order to recognise spoken words and translate them into text, speech recognition combines linguistics and computer science. It enables machines to comprehend spoken language. In order for a computer to process spoken words, speech recognition first converts them into text. The process typically involves several stages, including acoustic modeling, language modeling, and decoding.

- Acoustic modeling involves capturing and analyzing the sound waves of spoken words, while language modeling involves analyzing the grammar and syntax of the language being spoken. Decoding is the process of matching the acoustic and language models to identify the words being spoken.
- Voice recognition is the capacity of a machine to hear spoken words and recognise them. Python's speech recognition software can then be used to translate the spoken words into text, ask a question, or respond. Even some equipment can be programmed to react to these spoken words. With the use of computer programmes that accept input from the microphone, process it, and convert it into a useful form, you can perform voice recognition in Python.

One of the most significant advances in speech recognition technology has been the development of deep learning algorithms, which have improved the accuracy and reliability of speech recognition systems. As a result, speech recognition technology is becoming more widespread and is expected to play an increasingly important role in how we interact with technology in the future. Although speech recognition appears quite futuristic, it is already commonplace. You can speak out your question or the one you want help with on automated phone calls, and voice recognition is also used by your virtual assistants like Siri or Alexa to converse with you naturally.

3.1.4 NumPy:

The essential Python package for scientific computing is called NumPy (Numerical Python), particularly for mathematical operations and computations. It is an all-purpose package for processing arrays. It offers a multidimensional array object with outstanding speed as well as capabilities for interacting with these arrays. NumPy provides functions to create, manipulate, and operate on these arrays, including operations such as addition, subtraction, multiplication, division, dot products, and many more. It is the cornerstone Python module for scientific computing. It has a number of characteristics, including the following crucial ones:

- a strong object for an N-dimensional array
- Complex (broadcasting) operations
- C/C++ and Fortran code integration tools
- useful Fourier transform, random number, and linear algebra abilities

Overall, NumPy is a powerful and versatile library that provides a rich set of mathematical and scientific tools for Python programmers. It is widely used in scientific computing, machine learning, data analysis, and other areas of research and development where numerical computation is required.

3.1.5 OpenCV:

A free and open-source software library for computer vision and machine learning is called OpenCV. For processing images and videos, OpenCV offers a variety of functions and algorithms, object detection and recognition, feature extraction, camera calibration, and 3D reconstruction.A standard infrastructure for computer vision applications was created with OpenCV in order to speed up the incorporation of artificial intelligence into products. OpenCV's BSD licensing makes it simple for companies to use and change the source. There's no need to install OpenCV individually because it comes packaged with Python bindings and dependencies. It supports Windows, Linux, Android, and Mac OS and offers C++, Python, Java, and MATLAB interfaces. OpenCV is designed to be user-friendly, with an easy-to-use API that allows developers to quickly and easily integrate computer vision functionality into their applications. Moreover, a variety of previously trained models for jobs like face detection and recognition are included. which can be used to build more complex applications with minimal effort.

3.1.8 MySQL Connector:

A MySQL connector is a software library or module that allows applications to communicate with a MySQL database. It acts as a bridge between an application and a database, enabling the application to retrieve, insert, update, and delete data from the database. A MySQL connector usually provides a set of programming interfaces and functions that allow an application to interact with a MySQL database, such as establishing a connection, executing SQL statements, and handling database errors. Python programmes can access MySQL databases with MySQL Connector/Python by using an API complying with the Python Database API Standard v2.0 (PEP 249). A version of the X DevAPI, an Application Programming Interface for interacting with the MySQL Document Store, is also included. These connectors are typically open-source and freely available for developers to use in their applications. Using a MySQL connector can simplify the development process and improve the performance and scalability of applications that need to interact with a MySQL database

3.2 Functional and Non-Functional Requirements:

3.2.1 Functional Requirements:

Functional means providing a particular service to the user and also providing specific behavior of the system. The functional output of the given service can be split into three main segments. The first and the primary segment is the Create Database segment whereby the database is created using the API call to YouTube, which in turn keeps updating itself from time to time in order to give results real time. There is a search term field which is required to take the user input. There is a visualization functionality which is used to obtain the graphs and the charts. There is a final results tab which helps us showcase the results.

API Request

- **Client Responsibilities**

1.1 The client will inquire about the process by sending a request to the API.
1.2 The client will spell the account number which is registered in the bank.
1.3 The valid client will request for the required operation from the specified operations.
1.4 The client can request for the copy that will be able to proceed with the next procedure.

- **Server Responsibilities**

2.1 The server will be activated when the client requests and explains the procedure.
2.2 The Server will validate the account number spelled by the client by comparing it with data from the database.
2.3 The operations requested by the client are processed and updates are made.
2.4 The server will generate the form according to the details from the client.

3.2.2 Nonfunctional Requirements

- **Performance requirements**

Our project is thought to be built on client-server architecture, meaning that the server should react if the client makes a request. Also, as the number of clients are going to be larger, the server gets overloaded. In order to serve all the requests of the clients there should be proper hardware and software networking capability. The architecture of the network should be such that the request/response time is measured. And the time between request and the response should be as minimum as possible.

- **Software quality attributes**

1. Adaptability: The software must be able to incur changes within itself as well as changes in the machine architecture.
2.Correctness: 20 The system should produce correct and exact output so as to evaluate the results.

- **Safety Requirements**

Other safety requirements should be that of power supply which should be uninterrupted. The networking devices should be properly connected. And faulty networking devices such as the router, switch and the hub should be removed as early as possible.

- **Security Requirements**

IDs and Passwords should be regulated to only certain lengths and the password must be strong, containing non-alphanumeric characters as well. The database of the user should be kept safe so that no one misses it.

3.3 REQUIREMENTS GATHERING

The project focuses on the characteristics of precision answering of questions asked by the user. Since most of them are only provided theoretically in the form of research papers, we intend to use them for practical applications. We use three methods for demand collection analysis:

1. Brainstorming

This is a way to gather ideas about a specific issue or problem from a team member. It helps identify all possibilities, simplifying the solution and prioritizing the best ideas originally requested. Therefore, after completing the documentation and observations, the team members discussed all the advantages and disadvantages of the existing system, as well as what should be strengthened and improved, in order to provide a better automation system. It also determines which methods can be implemented.

2. Observation

Determine the next improvement by studying the user, the system process and its shortcomings. It can be active or passive. When we want to understand each step of a process, we take the initiative to observe, and passively observe where we record and / or questionnaires, and wait for the work to be completed in order to complete the work without interruption. In this project, we choose passive observation, because the best interests of farmers are to let them complete the work without any interference, and then clear the doubts in order to make a better analysis of the improvements needed.

3. Literature analysis

This is one of the most important technologies. It relates to the written documentation of the system, the work and procedural methods it uses. In addition, these can help determine the requirements for the current project start. It also helps to validate and validate the method and determine which key points have been missed in the previous system, and should consider using the upcoming technology to create a new system.

3.4 System Requirement

3.4.1 Software configuration

Windows 7 or newer, 64-bit macOS 10.13 or later, or Linux, such as Ubuntu, RedHat, CentOS 6 or later, are all acceptable operating systems.
Linux-64 x 86, 64-bit Power8/Power9, MacOS-64 x 86, Windows-64 x 86, and 32-bit x86 make up the system architecture.
Environment: Python version 3.4.2 or higher, minimum 2.7

3.4.2 Hardware Configuration

Intel core i5 or above RAM of at least 8GB
Storage: A minimum of 5GB of disc space is needed for installation.

CHAPTER 4

Chapter 4
SYSTEM STUDY

4.1 Feasibility study

The proposal is viable, according to examination during the system study phase. Cost projections are developed for the project's overall strategy. During the system analysis, the proposed system's viability is considered. This will lighten the company's load and ensure everyone's
safety. For the goal of conducting a feasibility study, it is crucial to comprehend the system's primary requirements.
Three essential factors in feasibility study are

4.1.1 ECONOMICAL FEASIBILITY

The goal of this analysis is to ascertain the system's potential financial impact on the company. The corporation has limited resources for the technology's research and development. Data must be used to support the expenses. Because much of the technology was available to the public, the intended system could be finished on time and within budget. Only specialist items needed to be purchased.

4.1.2 TECHNICAL FEASIBILITY

The purpose of this study is to evaluate the system's technical requirements or viability. Every system created must not place an undue burden on the technical resources at hand. In light of this, there will be significant pressure on the existing technological resources. As a result, there will be high expectations for the client. Because it just requires minor or no changes to be implemented, the built system must have a minimal requirement.

4.1.3 SOCIAL FEASIBILITY

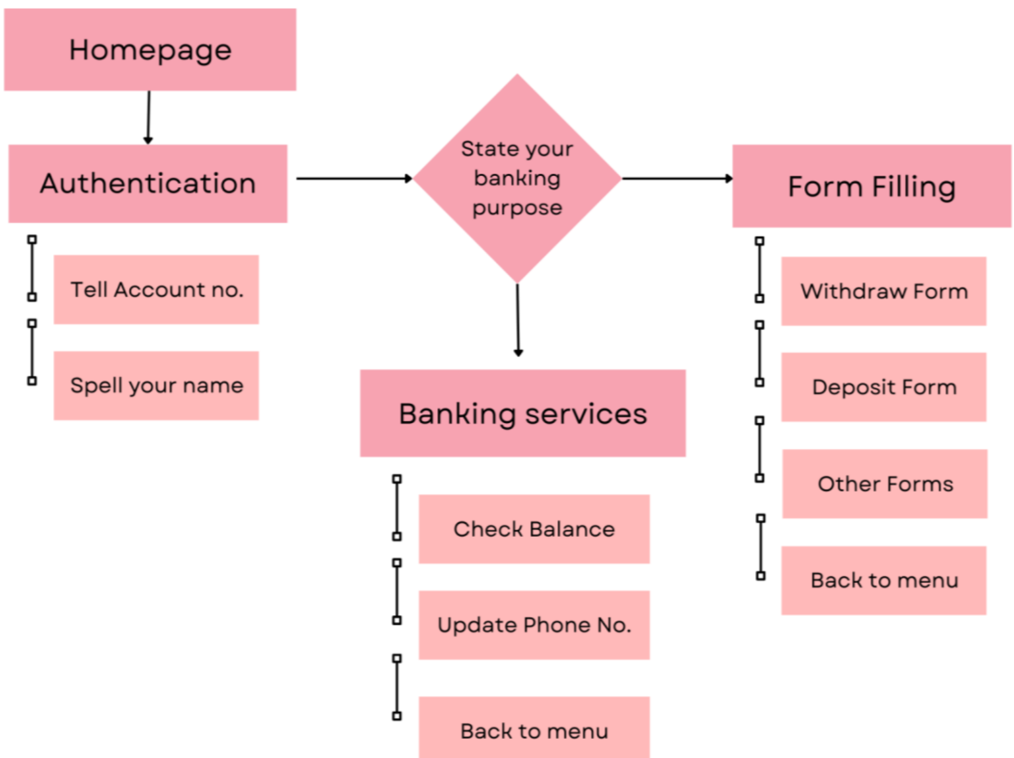
Finding out how much the user accepts the system is the aim of the study. Included in this is the instruction needed for the user to operate the system effectively. The system should not be seen by the user as a threat, but rather as a requirement. The system's level of user acceptability is only influenced by the methods employed to inform and familiarize them with it. His confidence needs to grow because he is the system's primary user and it is urged that he provide some constructive comments.

CHAPTER 5

Chapter 5
FLOWCHART AND USE CASE DIAGRAMS

Flowchart:

A flowchart is a visual representation of a program or process that uses lines and symbols to illustrate the different steps. It is used to show how steps and decision-making criteria move through a process, making the reasoning of a process easy to understand and articulate.



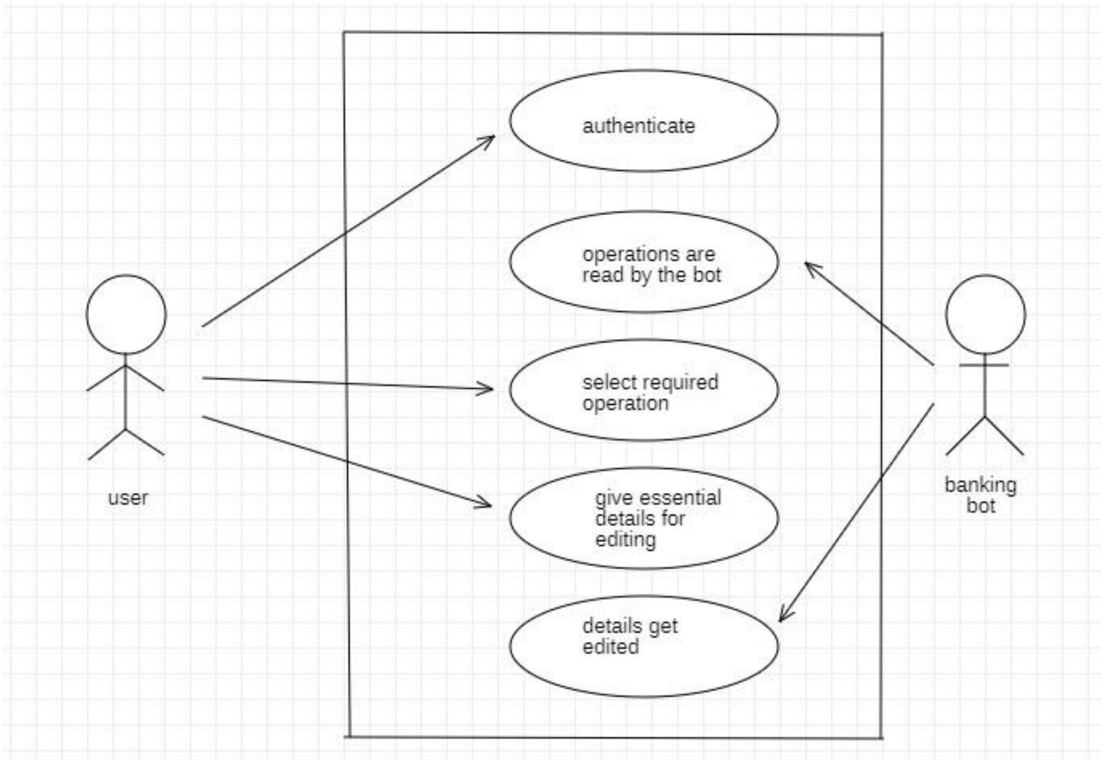
5.1 Flowchart: *Representing the flow of system*

Use case diagram :

The UML behavior of a system is shown in use-case diagrams, which also help in recording system specifications. Use-case maps provide an overview of a system's capabilities and high-level operations. These diagrams also show the interplay between the system and its players.

We have two main actors in our system:

- User: Mainly responsible for giving their required information and utilizing the services of the model.
- Banking bot: Its responsibility is to provide the services to the user like form fillings and updation of their information by recognizing their voice.



5.2 Use Case Diagram: *Represents the behavior of the system*

Activity diagram:

A sequence of activities is displayed in an activity diagram to show how an activity flows. The system description includes an activity map, which is crucial. The deed or system function that is being described.

Activity Diagram Notations -

- *Initial State* **-** The initial state is used to represent the original situation in which an action begins.

- *Activity or Action* **-**An activity depicts the performance of an action on or by things.

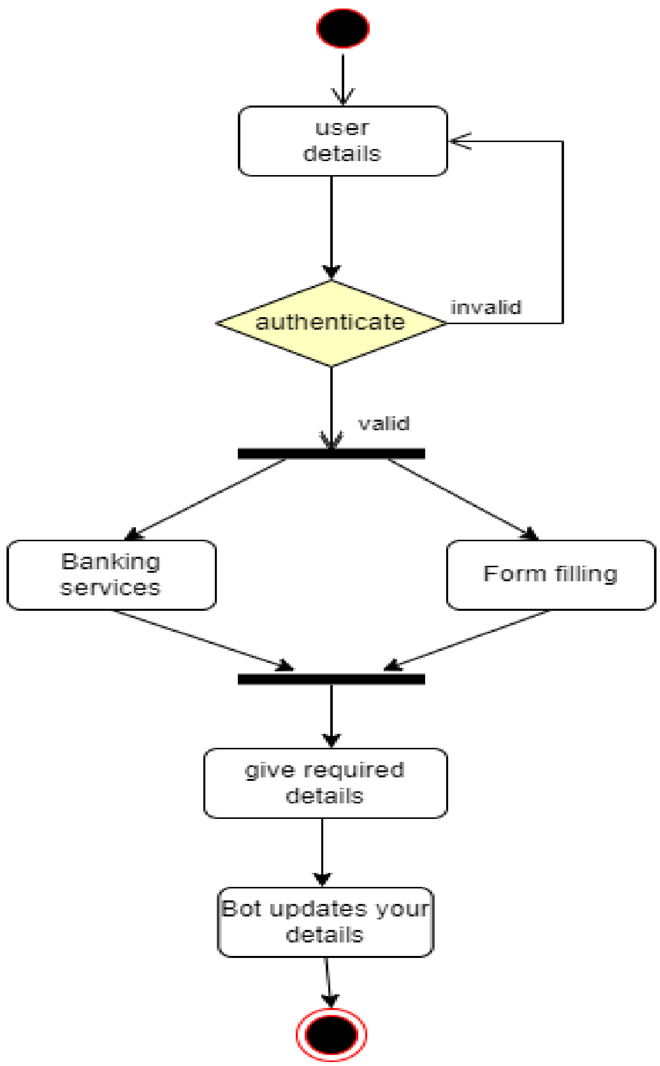
- *Action Flows or Control Flows* **-** Action flows or control flows are also known as paths and edges. They serve to visually depict the transition from one step of an action to another.

- *Decision node and Branching* **-** We use the judgment node when we need to make a choice before determining the direction of control.

- *Fork* - Concurrent actions are supported by fork nodes.

- *Join* - Join nodes are used to help several tasks come together as one. With join notations, there are two or more inbound edges as well as one outward edge.

- *Final State or End State* - A Final condition, also known as an End State, is the condition that the system achieves when a specific procedure or action is completed. The ultimate state in a state machine diagram is denoted by a full circle enclosed by another circle. There are various possible final stages for a system or procedure.



5.3 Activity Diagram: *Representing the activities of the system*

CHAPTER 6

Chapter 6
METHODOLOGY

6.1 MAIN TYPES OF CHATBOTS

6.1.1 Rule-Based Chatbots:

A rule-based chatbot is very simple. You are given a database of answers and a set of rules to help you find the right answer from the provided database. You can't create your own answers, but with a large database of answers and well-designed rules, they can be very effective and useful. The simplest form of rule-based chatbot has an array of individual inputs and their responses. These bots are very limited and can only respond to requests with exact matches to entries defined in the database.

6.1.2 AI-Based Chatbots

With the increasing use of machine learning in recent years, new approaches to building chatbots have emerged. Thanks to artificial intelligence, it has become possible to create highly intuitive and precise chatbots that are tailored to specific goals.Unlike rule-based types, AI-based chatbots can learn on their own because they rely on complex machine learning models. Now that we know how chatbots work, let's take a look at the libraries used to build simple rule-based chatbots. We are going to design a Conversational AI chatbot which is a combination of both mentioned above.
To build a Conversational AI chatbot we need to use RASA Framework.

6.2 ALGORITHMS USED

6.2.1 Speech_recognition

Python's speech recognition module does voice recognition using a variety of algorithms and methods. The speech recognition module employs a number of important algorithms and methods, including:

- HMMs, or hidden Markov models: These are statistical models that are used to simulate the probability distribution of speech sounds and sound sequences. Based on their acoustic properties, these models are used to identify spoken words and phrases.
- Gaussian Mixture Models (GMMs): To represent the probability distribution of speech sounds and sequences of sounds, GMMs are combined with HMMs.

- Neural Networks: Using neural networks, it is possible to identify spoken words and phrases based on their auditory properties. The speech recognition module recognises voice using neural networks that have already been trained.
- Language models: By taking into consideration the likelihood of particular word sequences and grammar rules, language models are used to increase the accuracy of speech recognition.

The speech recognition module uses these methods and techniques to do speech recognition, which enables it to accurately translate spoken words and phrases into text.

6.2.2 pyttsx3

Using a text-to-speech synthesis engine, the Python pyttsx3 module turns written text into spoken words. This engine generates speech using a variety of algorithms and methods. The pyttsx3 module employs a number of important algorithms and methods, such as:

- Concatenative synthesis: It is a technology that creates new speech by concatenating previously recorded voice segments. The objective is to progressively decipher a sentence's meaning or structure as it is being read, spoken, or written. rather than waiting for the whole sentence to be finished, or typed. Using this method, the pyttsx3 module creates speech by stringing together pre-recorded speech clips to create full phrases.
- Prosody modeling: Prosody modeling is a technique that entails simulating the rhythm, intonation, and stress of speech. This method is used by the pyttsx3 module to produce speech with realistic-sounding intonation and rhythm.

The pyttsx3 module can produce high-quality speech utilizing these algorithms and methods, making it a useful tool for developing speech-enabled applications.

6.2.3 FPDF

A well-known Python library for creating PDF files is called FPDF. In order to efficiently create and manipulate PDF files, it employs a variety of methods. The FPDF module employs a number of algorithms, including:

- Algorithms for page layout: FPDF employs algorithms to set up the content of each page in a PDF file. The dimensions and placement of each element on the page, including the text, pictures, and shapes, are determined by these algorithms.
- Font rendering algorithms: Algorithms for rendering text in different fonts and font styles are used by FPDF. Each character in the text is sized, spaced, and positioned according to these algorithms.

- Algorithms for image compression: FPDF employs techniques to compress images in a PDF file to minimize file size. These algorithms effectively compress photos while preserving their quality by using methods like JPEG compression.

FPDF is a potent tool for producing PDF documents in Python since it effectively creates and manipulates PDF files using a number of methods and techniques.

6.3 RASA Framework Working

6.3.1 Overview

The Rasa framework is made up of two components, Rasa NLU and Rasa Core. Rasa NLU is an open-source natural language processing engine for intent and entity classification. It basically processes unstructured user input and extracts meaning from it. There are many steps involved in this. Tokenization, vectorization, POS tagging, intent classification, etc. is a number of steps to name a number. Rasa Core is a machine learning framework that acts as a dialog management model. It takes structured input from the Rasa NLU and using a machine learning algorithm it predicts the output to display. Simply put, the Rasa NLU is used to interpret the user's unstructured input and convert it into structured data. Rasa core is used to decide the most likely response or action. Rasa NLU and Rasa Core are independent of each other and can be used separately if needed.

6.3.2 Understanding NLP(Natural Language Processing)

Natural language comprehension (NLU) and natural language generation (NLG) are two mechanisms that are combined in natural language processing (NLP). NLP is a superset of NLU and NLG. NLU understands the importance of user input. Focusing primarily on machine readability, his NLU allows chatbots to understand the meaning of text. NLU basically understands the text input given to it and classifies it into appropriate content.

NLU includes:

- Natural Language Inference(NLI) and Interpretation
- Semantic analysis
- Answer the questions
- Sentiment analysis
- Summary of user comments

NLG is software that generates human-understandable texts. Ideas for creating systems that can utilize the skills and expertise of people and computers are provided by NLG techniques.

NLG involves:

- Content identification
- Discourse planning
- Sentence aggregation
- Lexicalisation
- Referring to expression generation .

Natural language understanding (NLU) and natural language output make up an NLP system (NLG). The NLU structure understands the user's purpose or context when text is entered into the NLP engine, and the NLG then generates the appropriate response.

CHAPTER 7

Chapter 7
SAMPLE CODE

```
import sys

# import pyjokes as pyjokes
import speech_recognition as sr
import pyttsx3
# import pywhatkit
import datetime
# import wikipedia
import pandas as pd
import numpy as np
from fpdf import FPDF
#
# pdf = FPDF()
# pdf.add_page()
# pdf.set_font('Arial', '', 14)

print(pd.__version__)
data = np.random.randint(0, 200, size=5)
# df = pd.DataFrame(data, columns=['column name'])
df = pd.read_csv('BANK.csv')

print(df.head())
listener = sr.Recognizer()
engine = pyttsx3.init()
voices = engine.getProperty('voices')
engine.setProperty('voice', voices[1].id)
engine.say('Welcome to Wisdom Banking Bot.')
# engine.say('tell your account number')

engine.runAndWait()
def talk(text):

    engine.say(text)
    engine.runAndWait()
    newVoiceRate = 90
    # engine.setProperty('rate', newVoiceRate)
```

```
def take_command():
    try:
        with sr.Microphone() as source:
            talk("listening bro")
            print('listening...')
            voice = listener.listen(source)
            command = listener.recognize_google(voice)
            command = command.lower()
            if 'wisdom' in command:
                command = command.replace('wisdom', '')
                print(command)
        print(command)
        return command
    except:
        print("didn't Get the Command")
        talk('did not Get the Command. want to restart?')
        talk('Say YES or No')
        response = take_command()
        print('this is the response',response)
        if('yes' in response or 's' in response):
            talk('restarting the system you have to give the operations from starting with
account number.')
            run_alexa()
        elif('no' in response):
            talk('closing the bot. thank you visit again.')
            sys.exit()
        else:
            talk('no response recorded closing the bot')
            sys.exit()
        # take_command()



def run_alexa():
    try:

        talk('tell your account number.')
        command = take_command()
        command=list(command)
        voice = ""
        for i in command:
            if i != " ":
```

```
            print(i,command)
            voice += i

    print(voice , type(voice))

    if voice.isnumeric():
        accountnumber = int(voice)
        talk("Your account Number is"+str(accountnumber))
        print("HELLO")
        talk("please say your Name")
        name = take_command()
        talk(name)
        name = name.capitalize()
        print(name,accountnumber)
        talk("Your name is"+name+"and your account number is"+str(accountnumber))
        found = df.loc[df['AccountNo'] == accountnumber]
        # print(found)
        details = list(found)
        print(details)
        if (found["AccountHoldersName"] == name).bool():
            print(found["AccountHoldersName"])
            talk("you are verified"+name)
            print("Authentication Success")
            ss = int(found["BALANCE AMOUNT"].values[0])
            print(ss)
            print(type(ss))
            # talk("Your balance is"+found["BALANCE AMOUNT"])
            talk("you can go on with your banking")
            talk("speak ONE OR Banking for banking Services, Two or fill form for form
filling")
            operation = take_command()
            print(operation)
            if('one' in operation or '1' in operation or 'bank' in operation):
                print("Operations executed")
                talk("speak one for balance checking , two for phone number update, three to
go to back menu")
                innerOper = take_command()
                print(innerOper)
                if ('one' in innerOper or '1' in innerOper or 'balance' in innerOper):
                    talk("your balance is "+str(ss))
                    print("your balance is"+str(ss))
                    print("Inner oper executed")
                    talk('you are going to starting of the operations')
```

```
    elif ('two' in innerOper or '2' in innerOper or 'number' in innerOper):
        talk("tell your Phone Number digit by digit like 1,2,3")
        phone = take_command()
        phone = list(phone)
        number = ""
        for i in phone:
            if i != " ":
                print(i, phone)
                number += i

        print(number, type(number))

        if number.isnumeric():
            phonenumber = int(number)
            # # data = df.loc[df['AccountNo'] == 1234560124]
            # indexx = df.loc[df['AccountNo'] == accountnumber].index
            # print("index of data = ", int(indexx.values[0]))
            # hehe = data["BALANCE AMOUNT"]
            # # print(int(hehe.values[0]))
            # df.loc[indexx, 'PhoneNumber'] = phonenumber
            # print(df.loc[df['AccountNo'] == accountnumber])

            data = df.loc[df['AccountNo'] == accountnumber]
            indexx = df.loc[df['AccountNo'] == accountnumber].index
            print("index of data = ", int(indexx.values[0]))
            ind = int(indexx.values[0])
            hehe = data["BALANCE AMOUNT"]
            print(int(hehe.values[0]))
            print("joo")
            print(indexx)
            print("hii")
            df.loc[ind, 'PhoneNumber'] = phonenumber
            print(df.loc[df['AccountNo'] == accountnumber])
            df.to_csv("BANK.csv", index=False)
            # if(df.to_csv("BANK.csv", index=False)):
            print("Phone number Updated Successfully")
            talk("updated successfully.")
            talk("redirecting to home.")
    elif('3' in innerOper or 'three' in innerOper):
        sys.exit()
elif('2' in operation or 'two' in operation or 'form' in operation):
    print('operation2 is executing form filling')
    talk('form filling')
```

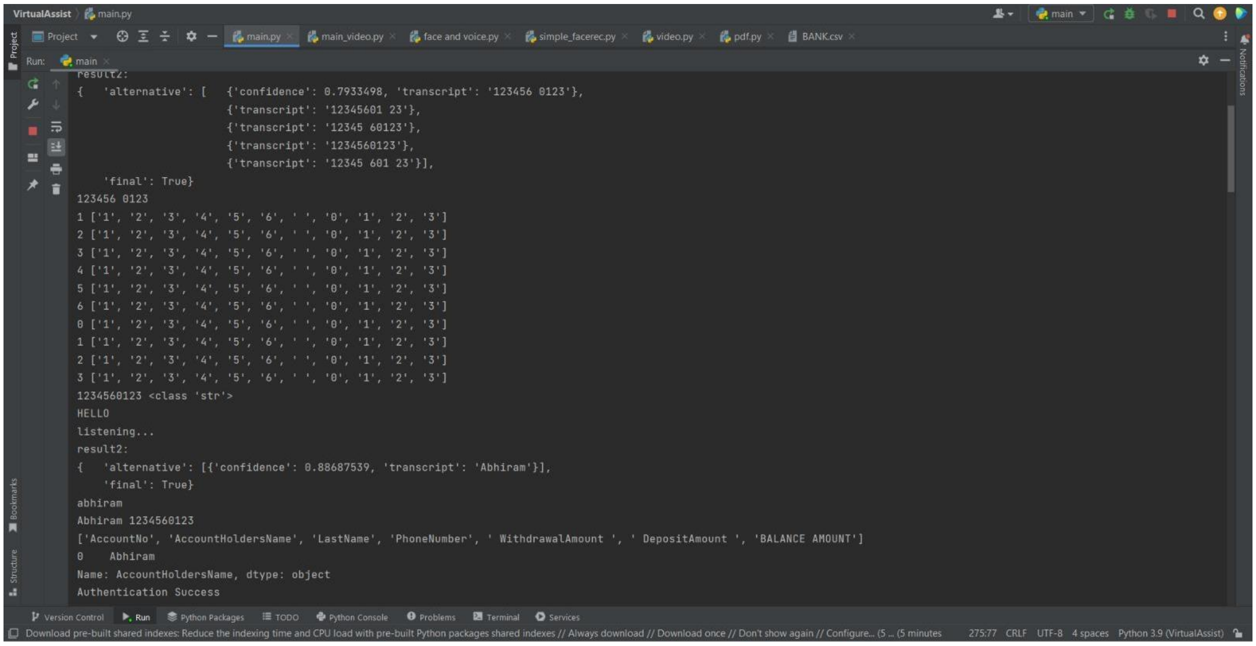
```
        talk('speak one or withdraw for withdraw form ,two or deposit for deposit
form ,three or exit for back menu')
        operation2 = take_command()
        if('one' in operation2 or '1' in operation2 or 'withdraw' in operation2):
            talk('tell the amount to withdraw from your account'+name)
            amount = take_command()
            amount = list(amount)
            cash = ""
            for i in amount:
                if i != " ":
                    print(i, amount)
                    cash += i

            print(cash, type(cash))

            if cash.isnumeric():
                cashAmount = int(cash)
                print(cashAmount)

                pdf = FPDF()
                pdf.add_page()
                pdf.set_font('Arial', '', 14)
                talk("you are withdrawing"+str(cashAmount))
                pdf.cell(200, 10, 'Bank for Blind\n', border=True, ln=5)
                pdf.cell(200, 10, 'WithDrawl Form\n', border=True, ln=5)
                pdf.cell(200, 10, 'Account No:\n'+str(accountnumber), border=True, ln=5)
                pdf.cell(200, 10, 'Account Holders Name:\n'+name, border=True, ln=5)
                pdf.cell(200, 10, 'Amount:\n' +str(cashAmount), border=True, ln=5)

                pdf.output('w'+str(accountnumber)+'g.pdf', 'F')
                talk('form filled successful')
                talk('you are going to starting of the operations')
        elif('two' in operation2 or '2' in operation2 or 'deposit' in operation2):

            talk('tell the amount to deposit in your account' + name)
            amount = take_command()
            amount = list(amount)
            cash = ""
            for i in amount:
                if i != " ":
                    print(i, amount)
                    cash += i
```

```
            print(cash, type(cash))

            if cash.isnumeric():
                cashAmount = int(cash)
                print(cashAmount)

                pdf = FPDF()
                pdf.add_page()
                pdf.set_font('Arial', '', 14)
                talk("you are depositing" + str(cashAmount))
                pdf.cell(200, 10, 'Bank for Blind\n', border=True, ln=5)
                pdf.cell(200, 10, 'Deposit Form\n', border=True, ln=5)
                pdf.cell(200, 10, 'Account No:\n' + str(accountnumber), border=True,
ln=5)
                pdf.cell(200, 10, 'Account Holders Name:\n' + name, border=True, ln=5)
                pdf.cell(200, 10, 'Amount:\n' + str(cashAmount), border=True, ln=5)

                pdf.output('D' + str(accountnumber) + 'g.pdf', 'F')
                talk('form filled successful.')
                talk('you are going to starting of the operations')
        else:
            print("bye bye")
            talk("Bye")
            sys.exit()

    else:
        print("Wrong input")
        talk('account number is not correct.')
        talk('restarting')
        run_alexa()

# elif 'play' in command:
#     song = command.replace('play', '')
#     talk('playing ' + song)
#     pywhatkit.playonyt(song)
# elif 'time' in command:
#     time = datetime.datetime.now().strftime('%I:%M %p')
#     talk('Current time is ' + time)
# elif 'who is' in command:
#     person = command.replace('who is', '')
#     info = wikipedia.summary(person, 1)
#     print(info)
#     talk(info)
```

```
        # elif 'work' in command:
        #
        #       print('I can work by taking the user text and converting to the machine understandable form, through Speech Recognition.And the working process is simple one can tell the required options through the speaker and i can fill the forms and give them through the printer ')
        #   talk('i can work by taking the user text and converting to the machine understandable form, through Speech Recognition.And the working process is simple one can tell the required options through the speaker and i can fill the forms and give them through the printer')
        # elif 'purpose' in command:
        #   print('I am wisdom banking bot. i am developed to solve the problems that are facing by the blind people in banking system. I can fill forms , i can tell the bank balance,and can talk multiple languages.')
        #    talk('I am wisdom banking bot. i am developed to solve the problems that are
facing by the blind people in banking system.')
        #    talk('I can fill forms , i can tell the bank balance,and can talk multiple languages')
        # elif 'operations' in command:
        #    print('one for form filling , two for banking services')
        #    talk('one for form filling , two for banking services')
        # elif 'joke' in command:
        #    talk(pyjokes.get_joke())
        elif 'close' in command:
            sys.exit()
        else:
            talk('Please say the command again.')
    except:
        talk('sorry system failed')
        talk("do you want to continue from starting?")
        talk("say yes or no.")
        response = take_command()
        if('yes' in response or 's' in response or 'restart' in response):
            run_alexa()
        elif('no' in response or 'close' in response or 'exit' in response):
            talk('closing the bot. thank you. visit again')
            sys.exit()


while True:
    run_alexa()
```

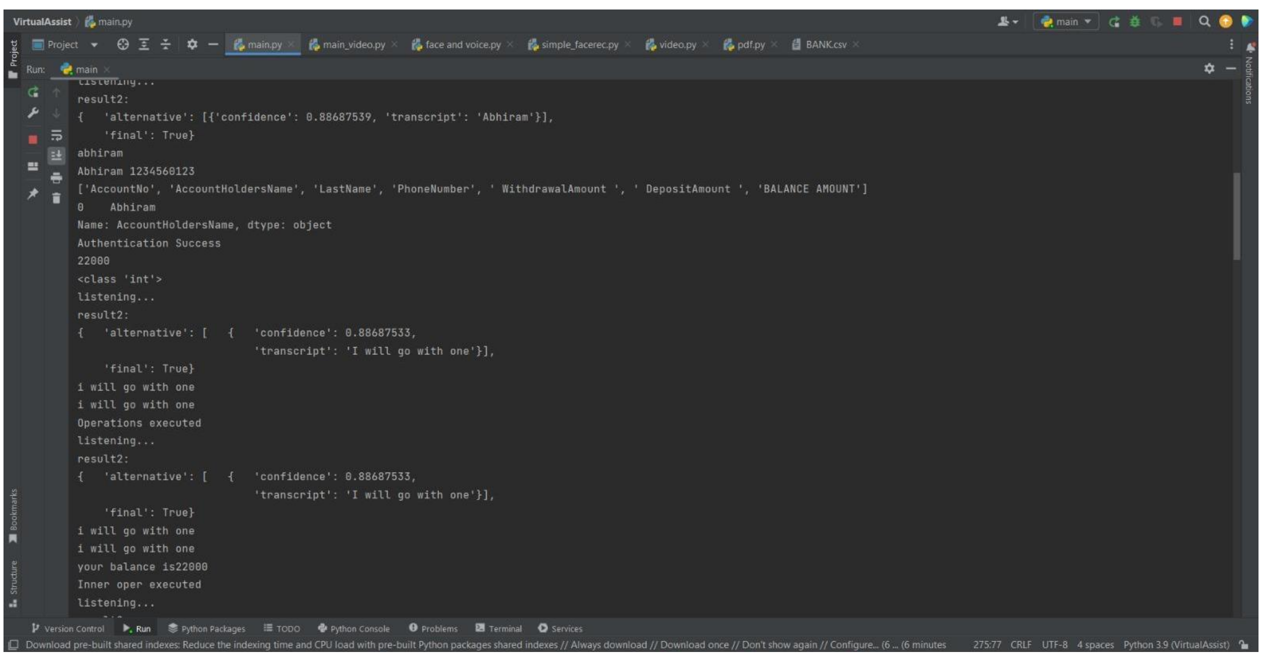
CHAPTER 8

Chapter 8
RESULTS AND ANALYSIS



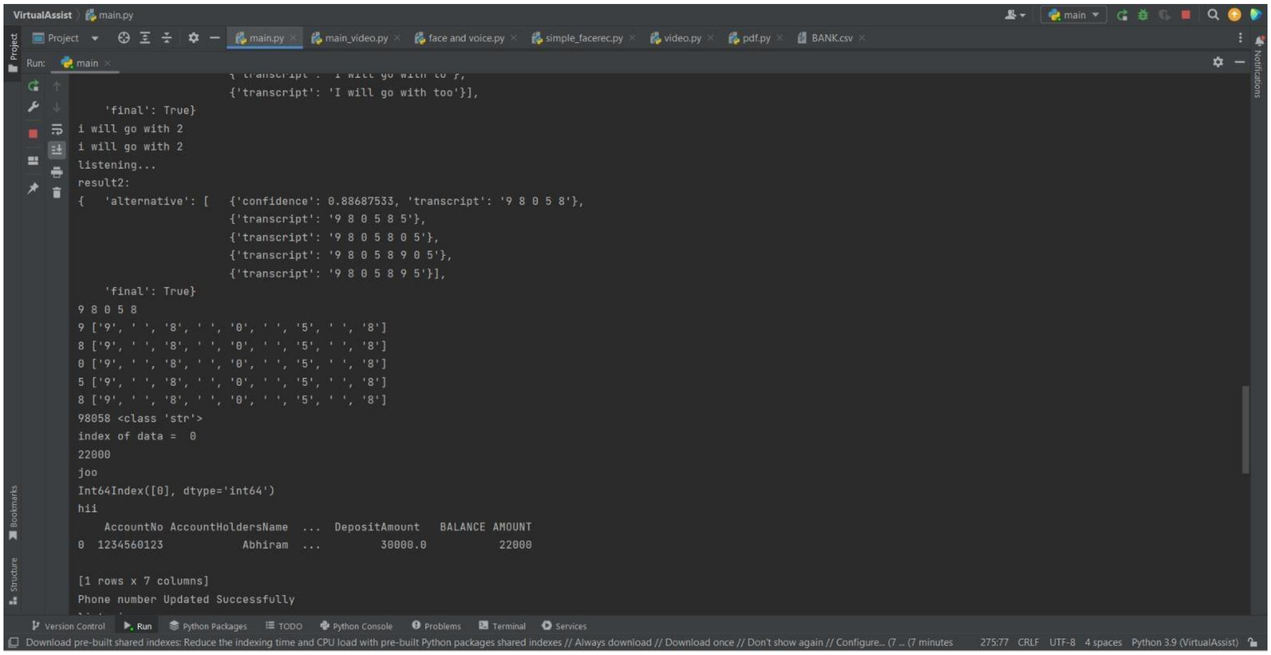
8.1 Authentication of the user

Authentication is done by checking if the account number matches their name.



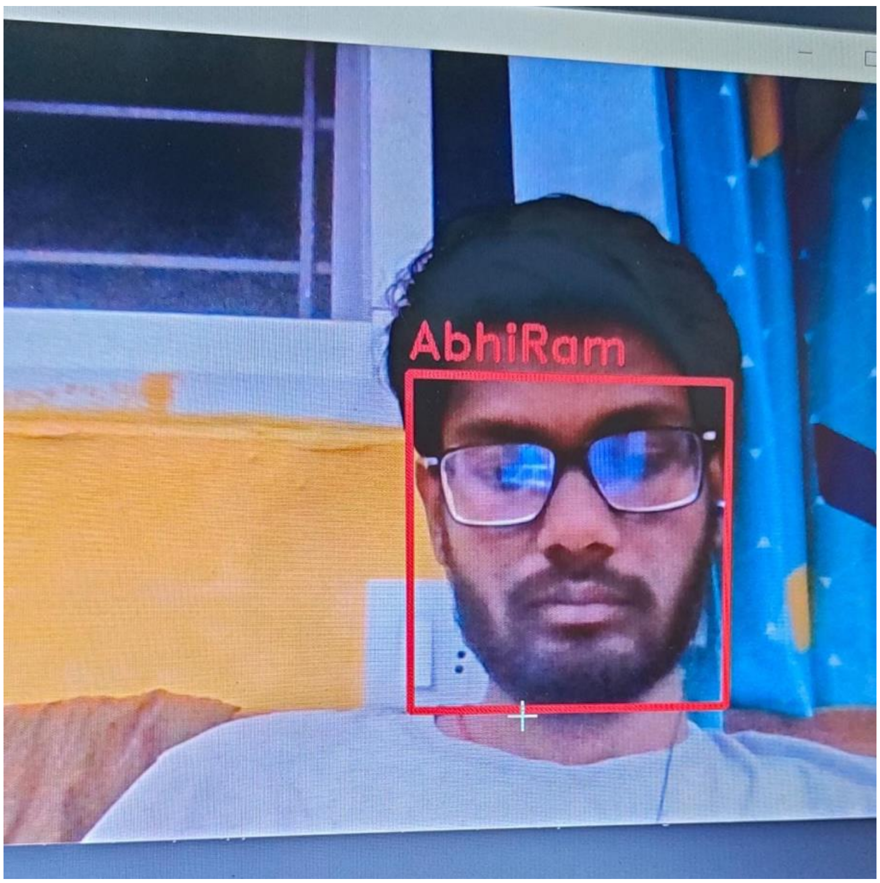
8.2 Balance Checking Option

On opting for option 1 the system gives the account balance as output.



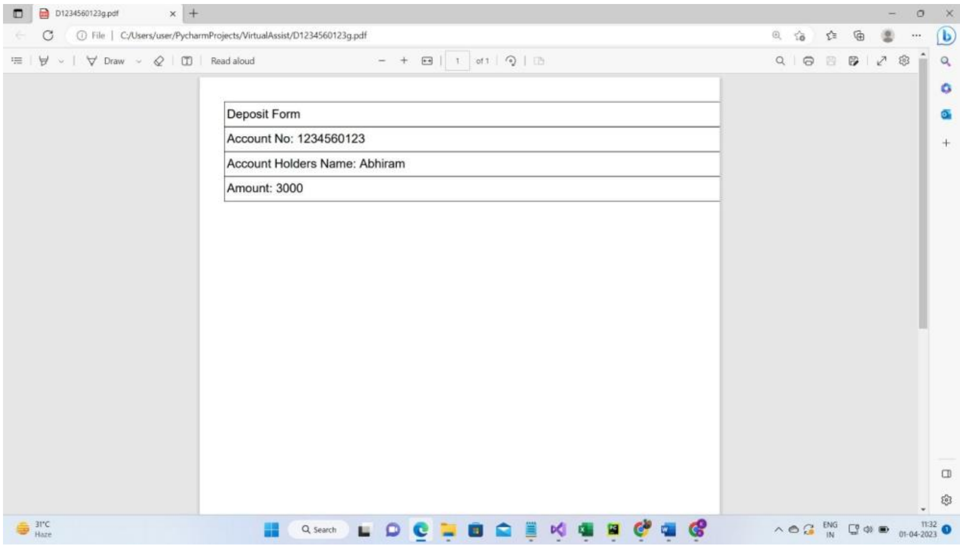
8.3 Phone Number Updation Option

On opting for option 2 the system performs phone number updation.

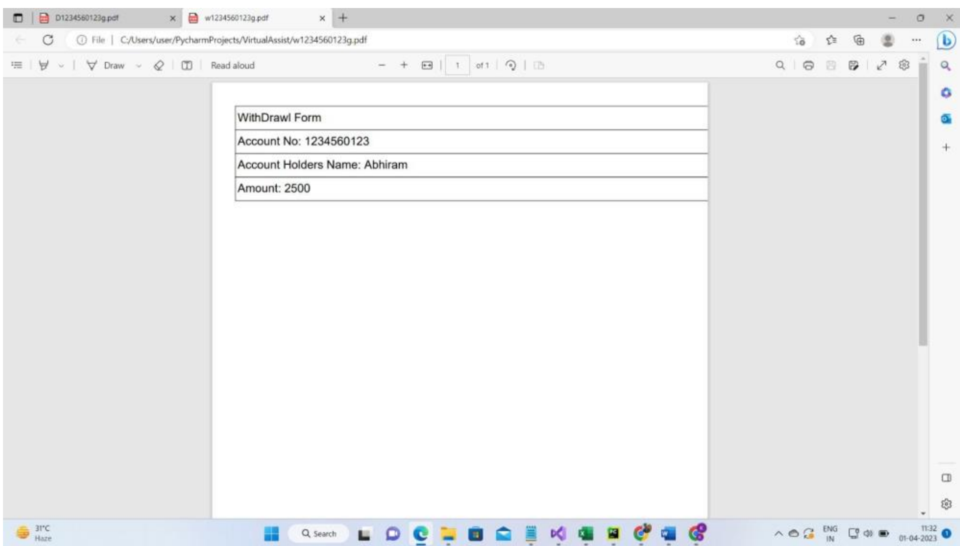


8.4 Face recognition

The system matches users using face recognition and unknown users can be found.



8.5 Example of a Deposit Form generated as an output.



8.6 Example of the Withdrawal From generated as an output.

CHAPTER 9

Chapter 9
CONCLUSION

Conclusion:

To conclude, Banking bot is helpful in guiding blind with correct and most up-to-date sources of information. It is advantageous for Visually impaired people for queries such as balance enquiry, Phone number update, filling various kinds of forms like withdraw from, deposit form. Customers can get the information at their fingertips rather than taking help from some other people. It improves efficiency by taking over tasks for which humans are not essential. This project talks about the capability of chatbots to help the blind people to access basic services of the bank.

Due to this, the conversation may become more enjoyable and intimate. They also only care about helping the people they interact with, which includes taking care of their needs by addressing inquiries and performing tasks. Applying the previously mentioned capabilities, we have added some basic chatbot features for banking services.

We are making our model using NLP, AI, Speech Recognition and ML algorithms that are useful for visually impaired people to access the bank on their own without any external help. chatbot that can interact with visually impaired users in a way that is accessible and user-friendly. Forms are filled by the chatbot by recognising the speech of the user.For this purpose we used open CV , speech recognition . Ai protected security is implemented.

Future work:

In today's digital age, more and more people are turning to digital solutions for their financial needs.

Generally blind people can access bank services but with the help of some trustworthy people. So,we developed these projects for some basic needs with security.In future we want to implement complete services of the bank . We want to include fingerprints for authentication purposes so there will not be any fraud activities . Language should not be a barrier in our life so we will update our project it can help people in all regions with different languages

CHAPTER 10

Chapter 10
REFERENCES

[1] A Smart Personal AI Assistant for Visually Impaired People by Shubham Melvin Felix, Sumer Kumar, and A. Veeramuthu Proceedings of the 2nd International Conference on Trends in Electronics and Informatics.

[2] Humorist Bot: Bringing Computational Humor in a Chat-Bot System by Augello A. Saccone G. Gaglio S. Pilato G., Humorist Bot: Bringing Computational Humor in a Chat-Bot System. Proceedings of the International Conference on “Complex, Intelligent and Software Intensive Systems (CISIS)”, 4-7 March 2008, Barcelona, Spain, pp.703- 708.

[3] Enabling Finger-Touch-Based Mobile User Authentication via Physical Vibrations on IoT Devices by Xin Yang, Student Member, IEEE, Song Yang, Student Member, IEEE, Jian Liu, Member, IEEE, Chen Wang, Yingying Chen, Fellow, IEEE, and Nitesh Saxena, Member, IEEE IEEE TRANSACTIONS ON MOBILE COMPUTING, 2020.

[4] Speech-based Virtual Travel Assistant for Visually Impaired by N. Sripriya, S.Poornima, S.Mohanavalli, R.Pooja Bhaiya , V. Nikita Department of Information Technology, SSN College of Engineering, Chennai, India.

[5] A Virtual Assistant for Visually Impaired Persons by Md.Rakibuz Sultan, Md Moinul Hoque, Department of CSE , Ahsanullah University of Science and Technology.

[6] Assistant for Visually Impaired using Computer Vision by Pratik Vyavahare, Syed Habeeb , Mechatronics Engineering Manipal University Dubai, UAE.

[7] Speech to text conversion for multilingual languages by Yogita H. Ghadage, Sushama D. Shelke, International Conference on Communication and Signal Processing, April 6-8, 2016, India.

PLAGIARISM AND PAPER

banking bot for blind

ORIGINALITY REPORT

12%	7%	3%	7%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to University of North Texas Student Paper	1%
2	github.com Internet Source	1%
3	Fred Torres-Cruz, Swati Tyagi, Manoj Sathe, S. Shyni Carmel Mary, Kapil Joshi, Surendra Kumar Shukla. "Evaluation of Performance of Artificial Intelligence System during Voice Recognition in Social Conversation", 2022 5th International Conference on Contemporary Computing and Informatics (IC3I), 2022 Publication	1%
4	kandi.openweaver.com Internet Source	1%
5	techprashantpokharkar.blogspot.com Internet Source	1%
6	Yogita H. Ghadage, Sushama D. Shelke. "Speech to text conversion for multilingual languages", 2016 International Conference on	1%

	Communication and Signal Processing (ICCSP), 2016 Publication	
7	Submitted to VNR Vignana Jyothi Institute of Engineering and Technology Student Paper	1%
8	Submitted to Vilnius Gediminas Technical University Student Paper	<1%
9	Submitted to Bangladesh University of Professionals Student Paper	<1%
10	www.classcentral.com Internet Source	<1%
11	Md. Rakibuz Sultan, Md Moinul Hoque, Farah Ulfath Heeya, Iftiquar Ahmed, Md. Redwanul Ferdouse, Shikder Mejbah Ahmed Mubin. "Adrisya Sahayak: A Bangla Virtual Assistant for Visually Impaired", 2021 2nd International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST), 2021 Publication	<1%
12	Submitted to The Manchester College Student Paper	<1%
13	www.coursehero.com Internet Source	<1%
	www.geeksforgeeks.org	

14	Internet Source	<1%
15	Submitted to University of Warwick Student Paper	<1%
16	docplayer.net Internet Source	<1%
17	ijres.org Internet Source	<1%
18	Amutha R, Mitha guru, Mohammed Nabeel Ahmed, Hemanth Rao Bhongale. "Chatbot using NLP", Advances in Intelligent Systems and Technologies, 2022 Publication	<1%
19	Submitted to University of Derby Student Paper	<1%
20	"Internet of Things, Smart Computing and Technology: A Roadmap Ahead", Springer Science and Business Media LLC, 2020 Publication	<1%
21	Submitted to National School of Business Management NSBM, Sri Lanka Student Paper	<1%
22	Submitted to University of Greenwich Student Paper	<1%
23	docs.anaconda.com Internet Source	<1%

24	www.ft.com Internet Source	<1%
25	www.sersc.org Internet Source	<1%
26	Mohammad Khorasani, Mohamed Abdou, Javier Hernández Fernández. "Chapter 4 Data Management and Visualization", Springer Science and Business Media LLC, 2022 Publication	<1%
27	orca.cf.ac.uk Internet Source	<1%
28	www.frontiersin.org Internet Source	<1%
29	www.textroad.com Internet Source	<1%
30	Submitted to University of Sydney Student Paper	<1%
31	mosis.eecs.utk.edu Internet Source	<1%
32	python.hotexamples.com Internet Source	<1%
33	www.reddit.com Internet Source	<1%
34	Submitted to Gitam University Student Paper	<1%

35	Krunal S. Trivedi. "Chapter 5 Fundamentals of Natural Language Processing", Springer Science and Business Media LLC, 2023 Publication	<1 %
36	archives.njit.edu Internet Source	<1 %
37	cse.anits.edu.in Internet Source	<1 %
38	devtales.tech Internet Source	<1 %
39	dspace.uiu.ac.bd:8080 Internet Source	<1 %
40	www.jetir.org Internet Source	<1 %
41	manualzz.com Internet Source	<1 %
42	myfik.unisza.edu.my Internet Source	<1 %
43	www.ijraset.com Internet Source	<1 %
44	"Applied Soft Computing and Communication Networks", Springer Science and Business Media LLC, 2021 Publication	<1 %

IJRAR.ORG E-ISSN: 2348-1269, P-ISSN: 2349-5138

INTERNATIONAL JOURNAL OF RESEARCH AND ANALYTICAL REVIEWS (IJRAR) | IJRAR.ORG

An International Open Access, Peer-reviewed, Refereed Journal

Banking AI Assistant For Unsighted People Using Natural Language Processing

[1]Pandit Samuel , [2]K Abhiram, [3]K Likhita, [4]M Eeshvar Gopal, [5]P Hanisha

[1]Assitant Professor, [2]Student, [3]Student, [4]Student, [5]Student
[1]Department of Information Technology,
[1]Anil Neerukonda Institute of Technology and Sciences, Visakhapatnam, India

Abstract: *The main objective of banking is to provide the services to the people to manage their money in a safe way. Banking is defined as "Accepting deposits of money from the public for the purpose of Lending or Investment, repayable on demand and withdrawable by cheque, draft, or otherwise. when it comes to the blind people they have to depend on others for daily activities. So, we are making an effort to help the blind in the banking sector which enables them to do physical activities like filling a form, check balance, update details etc on their own. People can use the banking services such as deposit, withdraw, loans, etc on their own. But the scenario when it comes to blind people he/she must have to depend on others to operate the bank account on behalf of the blind person. The bank may accede visually impaired customers request to appoint a person/persons as his/her Power of Attorney or Mandate Holder to operate his/her bank account. If the person is not the trusted one, then the blind people may lose their power on their account where he/she may lose the money or information which is private to his account. We have come up with an innovative solution to solve all the above-discussed problems.*

The Rule-based is the simple one when compared to other ones and can respond to customers based on predefined original, human-generated rules. Communication in this form tends to be text-based and often uses on-screen buttons, giving the user a wide choice of responses. These bots can also work by understanding certain keywords and phrases in written communication.

.

***Index Terms* – : Banking Bot ; Rule Based ; Artificial Intelligence ; Natural Language Processing .**

1. INTRODUCTION

Introducing our innovative banking bot, specifically designed to provide seamless banking experience to the visually impaired. This bot is equipped with state-of-the-art technologies that make it easy for users to access and manage their bank accounts without any assistance. The bot's intuitive interface, voice recognition and text-to-speech capabilities ensure that users can interact with it easily and efficiently. With our bot, blind users can perform a wide range of banking transactions, from checking account balances and updating phone number along with filling forms, all without the need for assistance from a third party.
Our banking bot is a game-changer for the visually impaired, empowering them with the ability to manage their finances with ease and independence.

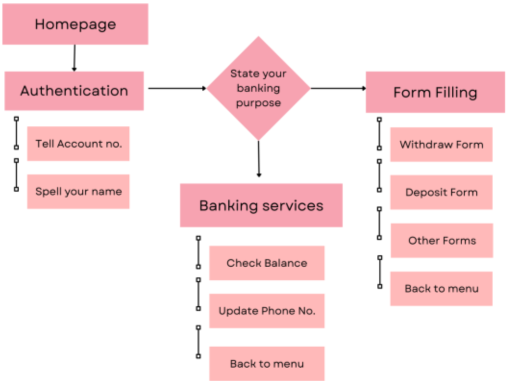


Figure-1 : Flow

The above Flowchart represents the flow of a banking transaction which starts from Homepage representing the Interface with the introduction of the Bot, then there will be the initial authentication of the user to login into the system with their account number. Then the user will be given their choice of choosing the operations of the Form Filling or other banking purposes. Finally the user's purpose will be fulfilled and the application returns to the Homepage.

2. Limitations of Traditional Banking Systems

Traditional methods of banking are generally considered but when it comes to the case of blind people , it's difficult for the transactions to be held .So we are introducing our banking bot to overcome the problem of trust issues.Hence enabling the online transactions for the Unsighted people will resolve the issue.

3.Need of the Banking Bot Application

Our Banking bot can be extremely helpful for visually impaired people, as they provide an accessible and convenient way to manage their finances independently. Here are some ways that a banking bot can help visually impaired people.

4. Technologies to be used in making Banking Bot

To succeed in making the bot different Python Libraries are to be used in different sectors of the project such as:

1. pyttsx3
2. pyaudio
3. speechrecognition
4. OpenCV
5. RASA Algorithm.

These libraries are to be used for speech recognition, audio to text conversion etc.

5. Methodology

5.1 Types of Chatbots

• **Rule Based Chatbots**

One of the simplest chatbot is rule based chatbot.In these the database of answers and rules are given that help the bot to find the right answer from the given database.very effective and useful answers are created. we can not create our own answers ,but with a large database of answers and well-designed rules.The individual inputs and their responses are stored in the form of arrays in these simplest rule-based chatbot.these bots are very limited and can only answer queries that exactly match the entries provided in the database.

• **AI-Based Chatbots**

Machine learning is increasing a lot in our recent life ,many new approaches and techniques are developed . For create highly intuitive and precise chatbots artificial intelligence are used that are tailored to specific goals.Using complex machine learning models the AI-based chatbots are designed.

because of these complex machine learning model these chat bot as a capability of learn on their own.

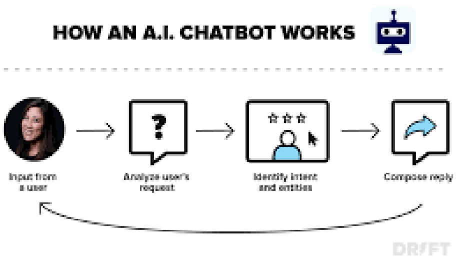


Figure 3:Chatbot Woking

5.2 RASA Framework Working

• **Overview**

The Rasa framework consists of two components: Rasa NLU and Rasa Core. Rasa NLU is an open source natural language processing engine for intent and entity classification. It basically processes unstructured user input and extracts the meaning from it. This requires many steps.

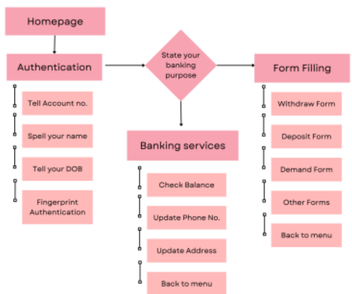


Figure : Working of the System

• **NLP(Natural Language Processing)**

The combination of the two processes natural language understanding (NLU) and natural language generation(NLG) the Natural language processing (NLP) is made.NLU and NLG are subsets of NLP.
the user input importance are understood by the NLU. Its NLU focuses primarily on machine readability, allowing chatbots to understand the meaning of text.The main work of the NLU is to understand the input text given by the users and it classifies it into appropriate content.
NLU includes:

- Natural Language Inference (NLI) and Interpretation
- Semantic analysis
- Answer the questions
- Sentiment analysis
- Summary of user comments

Human understandable texts are generated by these NLG software. NLG provide the techniques on how to build systems that can leverage the knowledge and capabilities of humans and machines.
NLG involves:

- Content identification
- Discourse planning
- Sentence aggregation
- Lexicalisation
- Referring expression generation

There are two components of an NLP system - natural language understanding (NLU) and natural language generation (NLG). When you enter text into the NLP engine, the user's meaning or context is decoded by the NLU structure and the response is generated by the NLG.

• **OpenCV**
OpenCV is a Python library that allows you to perform image processing and computer vision tasks. It provides a wide range of features, including object detection, face recognition, and tracking. OpenCV helps us to compare two finger prints whether they are same or not. It uses an algorithm known as SIFT (Scale-Invariant Feature Transform)

6. Capabilities of the Banking Bot

Voice recognition: Banking bots are equipped with voice recognition technology, which allows visually impaired users to interact with the bot using their voice. They can ask questions, give commands, and carry out banking services without needing to see a screen or type.

Text-to-speech capabilities: Banking bots can also read out the information on the screen, such as account balances and other account details, using text-to-speech capabilities.

Intuitive interface: Banking bots are designed with an intuitive interface that is easy to navigate using simple voice commands. This means that visually impaired users can perform a range of banking transactions.

Availability: Banking bots are available, which means that visually impaired users can access their accounts and perform banking transactions at any time.

6.1 The Role Of Text to Speech -: One of the implementations of the Banking Bot includes the conversion of the text to speech, where the Data of the bank is to be spoken out by the bot to help the user listen to the given options. pyttsx3 is a text-to-speech conversion library in Python. It works offline and is compatible with both Python 3. An application invokes the pyttsx3.init() factory function to get a reference to a pyttsx3 library. It is a very easy to use tool which converts the entered text into speech.

6.2 The Role Of Speech Recognition -: With the aid of speech recognition technology, a computer can comprehend spoken words. It is sometimes referred to as speech-to-text or automated speech recognition.Python's speech recognition software can then be used to translate the spoken words into text, ask a question, or respond. Even some equipment can be programmed to react to these spoken words.

6.3 The Role Of OpenCV-: OpenCV is a free and open-source software library for computer vision. As a part of security of the banking transactions, The Face of the User will be detected throughout the transaction processing, and the OpenCV library is used for the face detection.

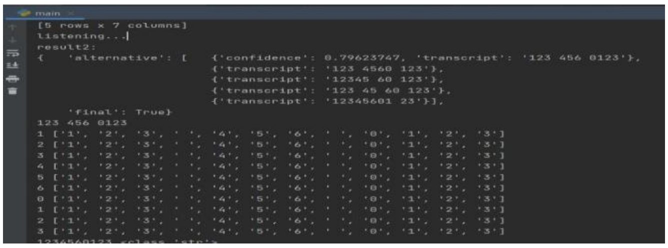
7. Evaluation Metrics

7.1 Metrics : The metrics to be considered for the application includes the calculation of the data validation, and the accuracy of the recovered data from the server.

$$Accuracy = \frac{Correct\ prediction}{Total\ cases} * 100\%$$

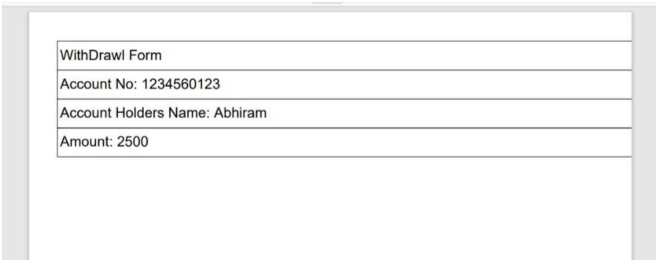
$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} * 100\%$$

7.2 Metrics Outcome : The accuracy of the system purely relies on the input given to the banking bot, so it may vary from person to person. Along with the accuracy of the bot, there will also be a calculation of the confidence of the system. This refers to the complete application. And unlike the accuracy, there is a calculation of the confidence of the application which will also vary from person to person.



7. Result and Experimental Analysis

We ran the experiments of our Banking Bot on PCs with Windows 11 Operating System,Intel Core i5-9500E @4.5GHz CPU with 8GB RAM. The utilization of the Banking Bot application has proved to be useful in terms of ease of banking, security throughout the transaction, and other similar banking features. The main functional category of the application is the Form Filling Feature which will be done with the complete security and also the output will be received as a PDF format and the minor features includes the updation of phone number, & also the Balance Checking is also included with the application.The security of the transaction includes the face recognition of the user. The face will be detected throughout the transaction whether it's a major or minor feature. If the face is not recognised there will be an error message displayed as Unknown User. This Security feature uses the pre registered face linked to that account. Hence the combination of most of the banking features including the required security features, results in The Banking Bot Application.


WithDrawl Form
Account No: 1234560123
Account Holders Name: Abhiram
Amount: 2500

Example Output for Withdrawal Form


Deposit Form
Account No: 1234560123
Account Holders Name: Abhiram
Amount: 3000

Example Output for Deposit Form

8. Scaling Features and hyperparameter frames

The Rasa framework is made up of the components Rasa Nlu and Rasa Core.For intent and entity classification these Rasa NLU is used as an open-source natural language processing engine. It basically processes unstructured user input and extracts meaning from it. There are many steps involved in this. Tokenization, vectorization, POS tagging, intent classification, etc. is a number of steps to name a number. Rasa Core is a machine learning framework that acts as a dialog management model. It takes structured input from the Rasa NLU and using a machine learning algorithm it predicts the output to display. Simply put, the Rasa NLU is used to interpret the user's unstructured input and convert it into structured data. Rasa core is used to decide the most likely response or action. Rasa NLU and Rasa Core are independent of each other and can be used separately if needed.

UnderstandingNLP(Natural Language Processing)
Natural language comprehension (NLU) and natural language generation (NLG) are two mechanisms that are combined in natural language processing (NLP). NLP is a superset of NLU and NLG. NLU understands the importance of user input. Focusing primarily on machine readability, his NLU allows chatbots to understand the meaning of text. NLU basically understands the text input given to it and classifies it into appropriate content.

NLU includes:

- NaturalLanguageInference(NLI)and Interpretation
- Semantic analysis
- Answer the questions
- Sentiment analysis
- Summary of user comments

NLG is software that generates human-understandable texts. Ideas for creating systems that can utilize the skills and expertise of people and computers are provided by NLG techniques.

NLG involves:

- Content identification
- Discourse planning
- Sentence aggregation
- Lexicalisation
- Referring to generation
- of expression

Natural language understanding (NLU) and natural language output make up an NLP system (NLG). The NLU structure understands the user's purpose or context when text is entered into the NLP engine, and the NLG then generates the appropriate response.

9. Conclusion

To conclude, Banking bot is helpful in guiding blind with correct and most up-to-date sources of information. It is advantageous for Visually impaired people for queries such as balance enquiry, Phone number update, filling various kinds of forms like withdraw from, deposit form. Customers can get the information at their fingertips rather than taking help from some other people. It improves efficiency by taking over tasks for which humans are not essential. This project talks about the capability of chatbots to help the blind people to access basic services of the bank.

This can give a feeling of interaction with humans and can make it a joyful conversation . By implementing the above functionalities, we have added some basic functions of chatbots for the banking services.

We are making our model using NLP, AI, Speech Recognition and ML algorithms that are useful for visually impaired people to access the bank on their own without any external help. chatbot that can interact with visually impaired users in a way that is accessible and user-friendly. Forms are filled by the chatbot by recognising the speech of the user.For this purpose we used open CV , speech recognition . Ai protected security is implemented.

10. Future Scope

In today's time , more and more functions are turning into a digital mode. Which also includes the time to time updation of the software is much required in terms of utility, and security. Which includes the future work as below.

1. Fingerprint Authentication at the initial stage of the transaction.
2. Availability of different languages included for regional convenience.

11. Authors' Note

The authors have disclosed that they have no conflicts of interest that may have influenced their study, and they have assured that their research work is authentic and does not contain any copied content.

12. References

[1] Shubham Melvin Felix, Sumer Kumar, and A. Veeramuthu Proceedings of the 2nd International Conference on Trends in Electronics and Informatics.

[2] Augello A. Saccone G. Gaglio S. Pilato G., Humorist Bot: Bringing Computational Humour in a Chat-Bot System. Proceedings of the International Conference on "Complex, Intelligent and Software Intensive Systems (CISIS)", 4-7 March 2008, Barcelona, Spain, pp.703-708.

[3] Xin Yang, Student Member, IEEE, Song Yang, Student Member, IEEE, Jian Liu, Member, IEEE, Chen Wang, Yingying Chen, Fellow, IEEE, and Nitesh Saxena, Member, IEEE IEEE TRANSACTIONS ON MOBILE COMPUTING, 2020.

[4] N. Sripriya, S.Poornima, S.Mohanavalli, R.Pooja Bhaiya , V. Nikita Department of Information Technology, SSN College of Engineering, Chennai, India..

[5] Md.Rakibuz Sultan, Md Moinul Hoque, Department of CSE , Ahsanullah University of Science and Technology.

[6] Pratik Vyavahare, Syed Habeeb , Mechatronics Engineering Manipal University Dubai, UAE.

[7] Yogita H. Ghadage, Sushama D. Shelke, International Conference on Communication and Signal Processing, April 6-8, 2016, India.

[8] MVladimir I. Ivanov and John S. Baras, Institute for Systems Research, University of Maryland, College Park, MD 20742, USA.

paper

ORIGINALITY REPORT

11%	7%	5%	8%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	www.ijirset.com Internet Source	2%
2	Gorli L Aruna Kumari, Poosapati Padmaja, Jaya G Suma. "A novel method for prediction of diabetes mellitus using deep convolutional neural network and long short-term memory", Indonesian Journal of Electrical Engineering and Computer Science, 2022 Publication	2%
3	Submitted to University of North Texas Student Paper	2%
4	Submitted to Johns Hopkins Unversity Student Paper	1%
5	bankingschool.co.in Internet Source	1%
6	Submitted to Visvesvaraya Technological University, Belagavi Student Paper	1%
7	www.ijitee.org Internet Source	<1%

8	Kondapalli Siva Prasad. "Unconstrained Optimization for Maximizing Ultimate Tensile Strength of Pulsed Current Micro Plasma Arc Welded Inconel 625 Sheets", Advances in Intelligent and Soft Computing, 2012 Publication	<1 %
9	"Global Trends in Information Systems and Software Applications", Springer Science and Business Media LLC, 2012 Publication	<1 %
10	Shirong Zhang, Bjorn B. de Koning, Fred Paas. "Effects of mouse pointing on learning from labeled and unlabeled split-attention materials: An eye-tracking study", Computers in Human Behavior, 2023 Publication	<1 %

Exclude quotes	On	Exclude matches	Off
Exclude bibliography	On		

Paper id: IJRAR_263203 – **Acceptance Notification and Review Result of paper** TITLE - BANKING AI ASSISTANT FOR UNSIGHTED PEOPLE USING NLP. - Your Paper Accepted Complete Below Process and Publish it. Your Email id: samuel.it@anits.edu.in Track your paper : https://ijrar.org/track.php?r_id=263203

WhatsApp 6354477117

editor@ijrar.org

ijrar.org

INTERNATIONAL JOURNAL OF RESEARCH AND ANALYTICAL REVIEWS (IJRAR.ORG) Impact Factor: 7.17 Calculated by Google Scholar

International Peer Reviewed & Refereed Journals, Open Access Journal

ISSN Approved Journal No: E-ISSN 2348-1269, P- ISSN 2349-5138 | ESTD Year: 2014

Scholarly open access journals, Peer-reviewed, and Refereed Journals, AI-Powered Research Tool , Multidisciplinary, Monthly, Indexing in all major database & Metadata, Citation Generator, Digital Object Identifier(DOI)

Your Paper Acceptance details as per below Complete Step 1 and Step 2 and publish within 1 to 2 days.

Dear Author, Congratulation!!!

Your manuscript with Registration ID: **IJRAR_ 263203** has been **accepted** for publication in the **International Journal of Research and Analytical Reviews (IJRAR) | www.ijrar.org** | **International Peer Reviewed & Refereed Journals, Open Access Journal. IJRAR is Peer Review Journal, Refereed Journal, Peer Reviewed Journal Referred Journal and Indexed Journal Open Access Journal Online and Print Journal**

IJRAR Impact Factor: 7.17 (Calculated by Google Scholar)

Check your paper status: **https://ijrar.org/track.php or http://ijrar.org/Authorhome/alogin.php**

Online Payment Link for Indian author: http://ijrar.org/pay_form_2.php

Online Payment Link for Foreign/international author: https://ijrar.org/pay_form_international.php

Your Paper Review Report :

Registration ID:	IJRAR – 263203
Title of the Paper:	**BANKING AI ASSISTANT FOR UNSIGHTED PEOPLE USING NLP.**
Accepted or Not:	**Accepted**

Criteria	**Points out of 100%**
Continuity	**93%**
Text structure	**88%**
References	**94%**
Understanding and	**92%**

Track Your Paper Details:	
Paper Track Link 1:	**https://ijrar.org/track.php**
Paper Track Link 2:	**http://ijrar.org/Authorhome/alogin.php**

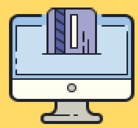
Online Payment link Details:

Illustrations		Indian author:	http://ijrar.org/pay_form_2.php
Explanatory power	91%	Foreign/international author:	https://ijrar.org/pay_form_international.php
Detailing	93%		
Relevance and practical advice	96%		

Overall Assessment (Comments.)	**Paper Accepted: YES**
Unique Contents: 89%	**Paper Accepted Complete Payment**

Reviewer comment store in online RMS system. Paper Accepted complete below Step1 and Step2 Process and Publish Your Paper within 1 to 2 day.

Publication of Paper	**Within 01-02 Days after Submitting documents. Please complete payment and send payment proof along with below documents.**



STEP 1: Pay Publication fees

For the pre-publication preparation, please complete the registration process as follow:

STEP 1: Pay Publication fees
(See following fees structure) *

Publication Fees and Payment Process	
Indian Author	**International Author (foreign)**
₹ 1500 INR.	**55 $ (USD)**

Pay Publication fees through online payment option only other option will take time for verification
Other payment option http://ijrar.org/paymentInfo.php

Preferred Payment link for Indian author:	**http://ijrar.org/pay_form_2.php (online payment Fast, Secure and best option of payment using this paper publish within 1 day)**
Preferred Payment link for Foreign/international author:	**https://ijrar.org/pay_form_international.php (online payment Fast, Secure and best option of payment using this paper publish within 1 day)**

*Charges include publication of one entire Research Paper Online, Individual Certificate for all authors (Softcopy), Confirmation Letter (Softcopy), Charges, Indexing, and Soft copy of Journal, typesetting, Paper formatting, maintenance of Website and journal infrastructures.

How to Pay Publication Fees?

You can See the details of Bank account and more option information through link http://ijrar.org/paymentInfo.php

International Author (foreign) Payment Option

By Online Payment for International Author (foreign) http://ijrar.org/paymentInfoforeign.php

Note: First use Preferred Payment link only if that not working then use alternative option 1,2,3,4,5.

You can pay fees through by your debit card / credit card / Net banking

Option	Remark
Preferred Payment link for Foreign/international author:	**https://ijrar.org/pay_form_international.php** **http://ijrar.org/paymentInfoforeign.php** **(online payment Fast, Secure and best option of payment using this paper publish within 1 day)**
Option 2 link :	**Payment link 2 Paypal Payment Link 3 for Payment** **Payment Link 4 pay 55 USD as equivalent to 4000 INR**

For Bank transfer follow this link only. If you use below bank account transfer than it will take time for verification and publication **http://ijrar.org/paymentInfoforeign.php**

Indian Author Payment Option	By Online Payment for Indian Author http://ijrar.org/paymentInfo.php	
	Note: First use Preferred Payment link only if that not working than use alternative option 1,2 and 3 You can pay fees through by your debit card / credit card / Net banking	
	Option	**Remark**
	Preferred Payment link for Indian author:	http://ijrar.org/pay_form_2.php (online payment Fast, Secure and best option of payment using this paper publish within 1 day)
	Option 1 link: UPI Payment like Paytm, bhim, googlepay app Note: Mentioned paper id in bank transfer and UPI without that we not consider payment or it will take delay in publication	Pay UPI payment through UPI app like BHIM app, google pay, PayTm UPI, BANK UPI etc. UPI id: 9898157864@upi, 9898157864@Paytm UPI QR Code : 9898157864@upi (Click Here and Open QR Code and Scan) UPI QR Code : 9898157864@paytm (Click Here and Open QR Code and Scan) Take the Receipt of Payment and send to editor@ijrar.org
	Option 2 link :	Payment link option 2

STEP 2: Send Payment Proof and Documents to editor@ijrar.org

If Publication fees paid, then complete below process after that publication will take 1 to 2 days.

Send payment screenshot to editor@ijrar.org

Please complete payment and send payment proof with below documents to email editor@ijrar.org only

STEP 2: Fill up Forms and Prepare Documents

Please complete payment and send payment proof with below documents

Visit your AUTHORHOME using your registration ID (Paper id IJRAR - 263203)) and email ID (Your registered Email ID – samuel.it@anits.edu.in) to complete below process.

Please Prepare following document and submit to editor@ijrar.org mention your paper registration ID in the subject.

Following forms is available on IJRAR website.
Login to Author Home http://www.ijrar.org/Authorhome/alogin.php or https://ijrar.org/track.php

1. **Final Formatted Manuscript/Research paper AS per IJRAR Paper Format** (must be in DOC file format and file name should be your registration ID)- Paper format is available on **IJRAR** website Download Paper format
2. **Undertaking by Author Form** (scanned copy)(Handwritten signed by author)(Download)
3. **Copy right transfer form** (scanned copy) (Handwritten signed by author)(Download)
4. **Identity Proof** of corresponding author only (scanned copy) (election Card/PAN Card/College Identity card/Driving License)
5. **Payment Proof** (Scan copy/Screen shot) also mentions Paid amount, Transaction ID, and Date of Payment in the mail whenever you send these documents.

If all author is not available than complete signature and identity proof process with single author. All author signature and identity proof not compulsory only one author signature and identity proof required.

Note: Please mention your Registration ID in the subject of mail whenever you communicate with us regarding any inquiry related to your research paper publication.

Publication of Paper	Within 01-05 Days after Submitting documents.
REQUEST FORM FOR HARD COPY OF CERTIFICATE	http://ijrar.org/paymentInfo.php

Thank you very much for submitting your article to the IJRAR.